

THE PROBLEM

When a controller commands a machine to run, a pulse-output sensor should send credible events. Short transitions may inflate a raw count; too few accepted events or a line held high can warrant review. These observations do not diagnose the physical cause.

WHAT THE CIRCUIT DOES

Synchronize a conditioned digital input; sample it with a clock enable; filter transitions using 1-8 consecutive stable samples; count accepted rising edges up to 255; snapshot each host-defined operating window with low-count, held-high, incomplete-transition and overflow observations. At 50 MHz, the default divider gives one sample per millisecond.

PIN CONTRACT

SIGNAL	FUNCTION
ui[0]	Raw conditioned digital pulse input
ui[3:1]	Stable samples minus one: 0-7 means 1-8 samples
ui[4], [5], [6]	Enable, synchronous clear, window boundary
ui[7]	Read live count or last completed count
uio_in[7:0]	Minimum expected pulses in the closing window
uo_out[7:0]	Selected saturating 8-bit count
uio_out[0:7]	Filtered, accepted, incomplete, low, overflow, ready, enable, held-high

LIVE PROOF

Five generated intended events and eight short transitions produce 13 raw edges. The actual RTL accepts five with a three-sample filter. The browser also demonstrates missing response, whole-window high input, and genuine pulses missed by strong filtering. The demo accelerates the divider to four clock cycles; it is not a sensor measurement. Local Icarus self-check and demo API tests pass.

ENGINEERING LIMITS

Filtering can lose genuine narrow pulses; the fastest setting can accept short changes. The count saturates at 255, and an incomplete-transition flag cannot catch changes entirely between sample ticks. A physical system needs input protection, clock, power and host control. 1x1 tile fit and 50 MHz remain targets until actual LibreLane layout and timing reports pass. No silicon or certified measurement is claimed.

SIGNAL PATH

